Dose-Aware MR-to-Density Synthesis through a Differentiable Pencil-Beam-Prior Residual 3D U-Net for Proton Dose Prediction on 0.35 T MRI (DoseRAD2026 Proton-MRI Task)

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Abstract. MR-only proton dose prediction is dominated by range: an sCT error that is dosimetrically benign for photons can displace the Bragg peak. We couple synthesis and dose in one differentiable pipeline: a classifier–refiner synthesizer maps the 0.35 T MR *directly to density* (bypassing the ambiguous HU inversion); differentiable physics channels; skin-entry water-equivalent path length (WEPL) and a self-contained GPU Hong pencil-beam prior; feed a residual 3D U-Net; and the synthesizer trains jointly through the physics with the dose loss plus a WEPL-consistency term that supervises integrated along-beam density, i.e. range. Under 5-fold cross-validation on the 75 public patients the system reaches plan $\gamma 1\%/1\,\text{mm} = 87.4 \pm 1.9$. A cross-validated real-CT ceiling of 96.1 ± 0.9 isolates the synthesis cost at 8.7 ± 1.9 points, twice the photon-task cost; and a distal-edge metric shows it is a range effect: mean distal-R_{80} shift 2.4 mm (lung 2.8) versus 0.4 mm on real CT. On the hidden test set, MR intensity shift was the dominant failure mode; shift-augmented classification recovered +3.1 board gamma. The final full-data submission scored $\gamma = 79.34$, per-beam MAE 0.0247, DVH 5.09 at 188.8 s per job (later reduced to 110 s by a batched engine), within the 500 s gate. We argue the residual gap is limited by the density information available at 0.35 T rather than by network capacity.

Keywords: MR-only proton therapy · synthetic CT · proton range · dose-aware synthesis · differentiable physics

1 Introduction

MR-guided and MR-only proton workflows are emerging for superior soft-tissue targeting and to avoid CT-to-MR registration error; they require an MR-to-density path, and its accuracy cost is the central question for their viability. That cost is particle-specific: proton range integrates density along the whole beam path, so synthesis errors compound into Bragg-peak shifts, whereas photon dose is density-tolerant. Existing MR-only proton dosimetry synthesizes a CT (or stopping power) under an *image* objective [1,2,3,4,5,6] and computes

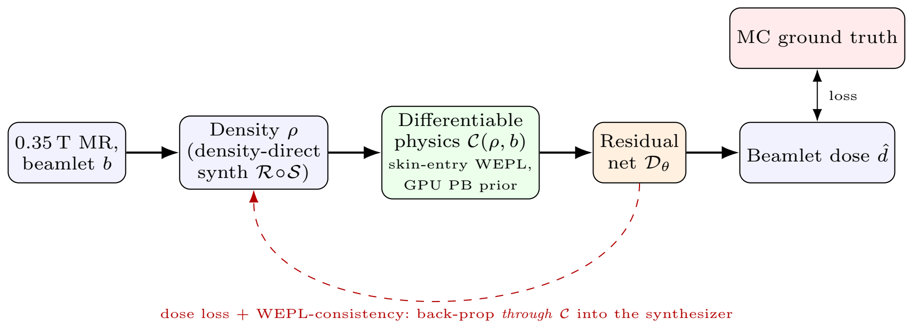

Fig. 1. Dose-aware MR-only proton pipeline: density-direct synthesis trained jointly through the differentiable physics with the dose loss and a WEPL-consistency (range) term.

dose separately. We instead make the synthesis *dose-aware*: an analytical, differentiable physics operator $\mathcal{C}$ (skin-entry WEPL, a GPU pencil-beam prior [7] supplying ∼95% of the dose, lateral distance, energy) feeds a residual 3D U-Net $\mathcal{D}_\theta$, and the MR-to-density synthesizer trains *through* $\mathcal{C}$ with the dose loss. Range receives two dedicated signals: the synthesizer outputs *density directly* (no HU round-trip), and a WEPL-consistency loss supervises the integrated along-beam density explicitly. This report details the Proton-MRI system and quantifies; via a cross-validated real-CT ceiling and a distal-edge metric, exactly how much accuracy MR-only proton calculation costs and why; the same framework serves our submissions to all four DoseRAD2026 tasks.

2 Methods

2.1 Data

Official DoseRAD2026 training set [10] (75 patients: 39 thorax, 36 abdomen; SynthRAD2025 cohort [11]): per patient a co-registered 0.35 T MR (ViewRay MRIdian, bSSFP) and planning CT, beam parameters, and per-beamlet MC dose-to-medium (matRad [9]) on the native 1×1×3 mm grid; several hundred to ∼a thousand beamlets per patient. The MRI-task ground truth is the CT-computed MC dose mapped onto the co-registered MR, so the real-CT dose is the achievable target (the framing of our ceiling analysis, Sec. 3.3). The sCT refiner additionally uses 166 field-matched 0.35 T MR–CT pairs from the public SynthRAD2025 pool (disjoint from validation); no other external data. Development: 5-fold CV over all 75; **the final submission is retrained on all 75**.

2.2 Model

Synthesizer $\mathcal{X} = \mathcal{R} \circ \mathcal{S}$: whole-image classifier $\mathcal{S}$ (air/lung/soft/bone → bulk coarse CT) and refiner $\mathcal{R}$; for protons the refiner outputs *relative density di-*

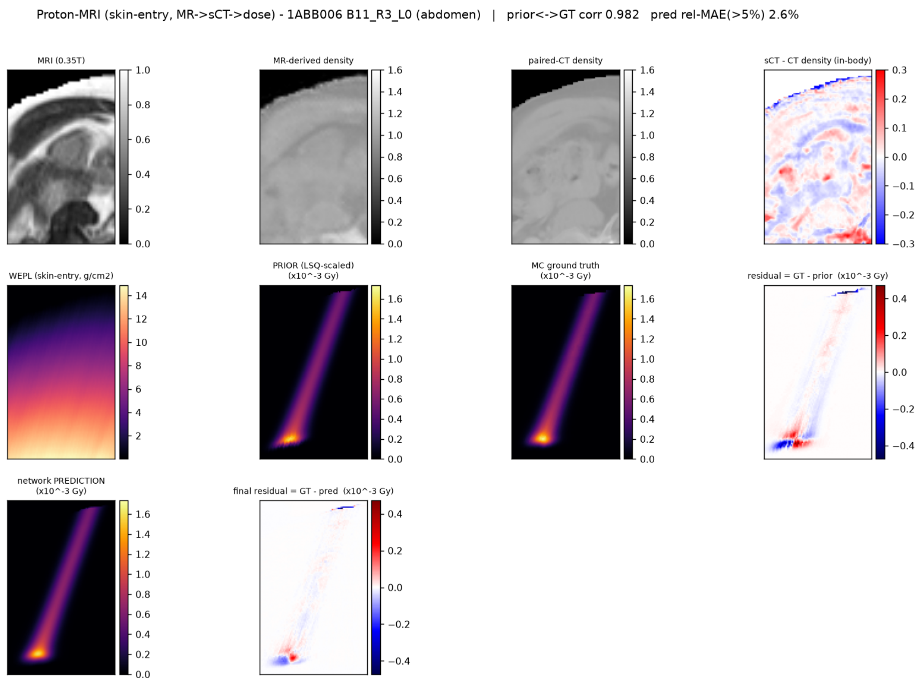

Fig. 2. Deployed Proton-MRI chain (abdomen, 1ABB006). Row 1: synthesis ; MR (0.35 T) → MR-derived density → paired-CT density → in-body density residual. Remaining panels: WEPL on the synthetic density → pencil-beam prior (Gy) → MC ground truth → residual (GT−prior) → prediction → final residual (GT−pred).

rectly (clamped to $[0, \rho_{\max}]$); the HU→density calibration is many-to-one and its inversion is an unnecessary, error-amplifying step between the network and the physics. The 2 mm synthesis is resampled to the native 1×1×3 mm dose grid by a differentiable physical-coordinate grid sample, so dose gradients reach the 2 mm synthesizer. **Physics** $\mathcal{C}$: five differentiable channels per beamlet; density/RSP, skin-entry WEPL (per-voxel, corrected scheme; the naive beam's-eye-view fan over-counts systematically), the GPU Hong pencil-beam prior (r=0.995 vs. pyRadPlan, ∼40 ms/beamlet, no planning-system dependency at inference), lateral distance, energy. **Dose network** $\mathcal{D}_\theta$: DoseUNet3D (base 48, ASPP bottleneck, Softplus head, FiLM conditioning), warm-started from Proton-CT. Figure 2 shows the deployed chain; synthesis and its dosimetric consequence in one figure.

2.3 Training

Stage 1: classifier/refiner pretraining (refiner also on the 166 field-matched pairs); dose network warm-started from Proton-CT. Stage 2 (dose-aware): refiner and dose network train jointly end-to-end (15–20k steps), the dose loss back-propagating

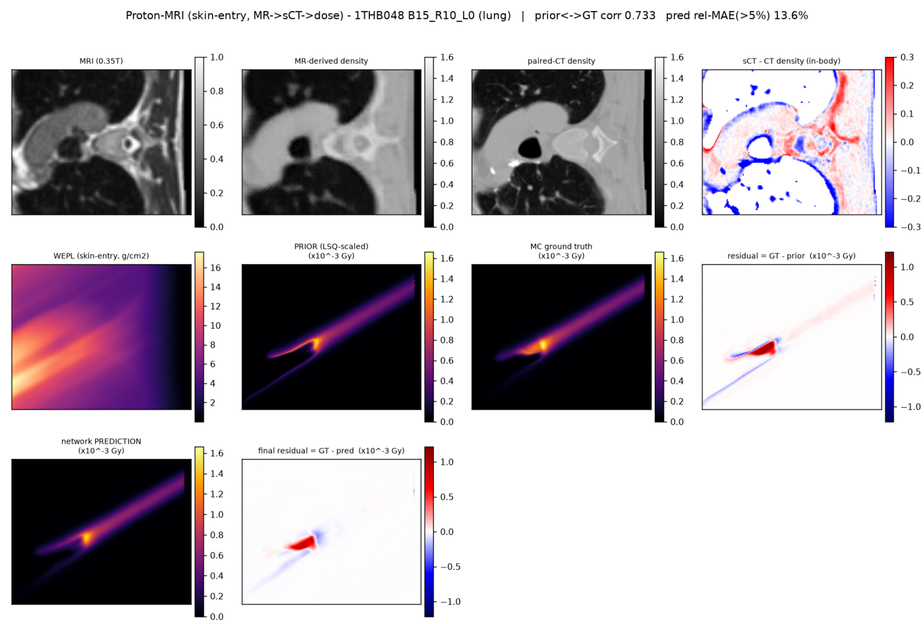

Fig. 3. As Fig. 2, lung case: the largest synthesis-induced range errors concentrate in lung, where 0.35 T MR carries the least density information.

through the WEPL and prior channels into the synthesizer, with two proton-specific additions: the *density-direct* output head and the *WEPL-consistency loss* $\lambda_w \|\mathrm{WEPL}(\hat{\rho}) - \mathrm{WEPL}(\rho_{\mathrm{CT}})\|_1$, which supervises the range-determining integral directly and is more localized than the dose loss. A region-weighted sCT anchor regularizes synthesis. **Hidden-test robustness:** leaderboard probes identified MR *intensity shift* as the dominant held-out failure mode, with the shared tissue classifier the main culprit; the deployed classifier is retrained with strong intensity-shift augmentation (recovering +3.1 board gamma), and, a deployment-critical detail; it must run *sliding-window* at the native grid: whole-image classifier inference collapsed lung on some test grids (single-patient gamma 90→48) before we gated for it. Reproducibility: fixed seeds, YAML configs, EMA checkpoints; code and weights to be open-sourced (Sec. 6).

2.4 Evaluation

Internal: plan-level local γ [8] (pymedphys, 1%/1 mm, ≥10% Rx cutoff, byte-identical to the vendored official code), per-beamlet masked MAE, official IDD and stratified-MAE reimplementations; frozen 16-patient held-out cohort gating every deployment change. Mechanistic probes: the *real-CT ceiling* (real CT substituted for the synthesis, patient-matched) and a *distal-edge range metric*; the absolute shift of the distal R_{80}/R_{90} along the beamlet axis on the 100

Table 1. Proton-MRI results. CV: 5-fold over the 75 public patients. Hidden test: official metrics of the final full-data submission (August 2026). Gate: 500 s/job.

Metric	Value
CV plan γ1%/1 mm (5-fold)	87.4 ± 1.9
CV real-CT ceiling / synthesis cost	96.1 ± 0.9 / 8.7 ± 1.9
Hidden-test per-beam masked MAE	0.0247 ± 0.0104
Hidden-test IDD curve distance	0.0233 ± 0.0101
Hidden-test stratified plan MAE	0.0253 ± 0.0098
Hidden-test plan γ1%/1 mm	79.34 ± 12.23
Hidden-test DVH clinical score	5.09 ± 5.57
Hidden-test runtime per job	188.8 s (batched engine: 110 s)

highest-dose beamlets per patient; because an aggregate gamma under-reports exactly the distal-range failure mode that matters for protons. Runtime: official $T = t_{\text{fix}} + N \cdot t_{\text{dose map}}$, gate 500 s/job, double-weighted.

3 Results

3.1 Accuracy: cross-validated and hidden-test

Per-fold CV $\gamma 1/1$: 84.0/88.9/88.9/86.7/88.6 (mean 87.4 ± 1.9). Development fold ($n = 16$): $\gamma 1/1 = 83.7 \pm 6.8$ (abdomen 88.8, lung 78.6), $\gamma 2/2 = 95.4$, $\gamma 3/3 = 98.4$; per-beamlet masked MAE $2.02 \pm 0.49\%$. Table 1 adds the hidden-test metrics. As on the photon-MRI task, the CV-to-hidden-test drop is dominated by MR intensity shift; the shift-augmented classifier recovered +3.1 board gamma over the unaugmented deployment and is part of the final submission.

3.2 The synthesis cost is a range effect

Re-running the trained pipeline of every fold with the *real* CT substituted for the synthesis gives a ceiling of 96.1 ± 0.9, closely reproducing our Proton-CT task accuracy (96.7 ± 1.0), so fed a real CT the MR pipeline behaves like the CT pipeline, and the deficit is genuinely a synthesis cost: 8.7 ± 1.9 points patient-matched, versus 4.7 ± 1.4 on the photon-MRI task ($\sim$1.9$\times$; on the development fold the paired contrast is -13.3 ± 6.1 vs. -5.0 ± 5.0, $p \ll 0.001$, with *every* patient losing under proton; Fig. 4). The distal-edge metric (Table 2) identifies the mechanism: the mean distal-R_{80} shift is 2.39 ± 1.04 mm on the synthesis versus 0.38 ± 0.20 mm on real CT, a $\sim$6$\times$ range error, concentrated in lung, exactly the failure mode an aggregate gamma under-reports.

3.3 Method-level ablations are small; the particle contrast is the robust result

Per-patient paired tests on the development fold (Table 3): the WEPL-consistency loss (+0.37, p=0.35) and density-direct synthesis (+0.36, p=0.25) are consistent,

Table 2. Distal-edge range accuracy: absolute shift of the distal R_{80}/R_{90} (mm) along the beamlet axis, 100 highest-dose beamlets per patient (development fold, 8 abdomen / 8 lung).

Input	Site	R_{80} shift (mm)	R_{90} shift (mm)
real CT	abdomen	0.25 ± 0.08	0.25 ± 0.07
real CT	lung	0.51 ± 0.20	0.47 ± 0.15
real CT	all	$\mathbf{0.38 \pm 0.20}$	0.36 ± 0.16
MR synthesis	abdomen	1.94 ± 0.53	1.94 ± 0.52
MR synthesis	lung	2.84 ± 1.25	2.59 ± 1.17
MR synthesis	all	$\mathbf{2.39 \pm 1.04}$	2.26 ± 0.93

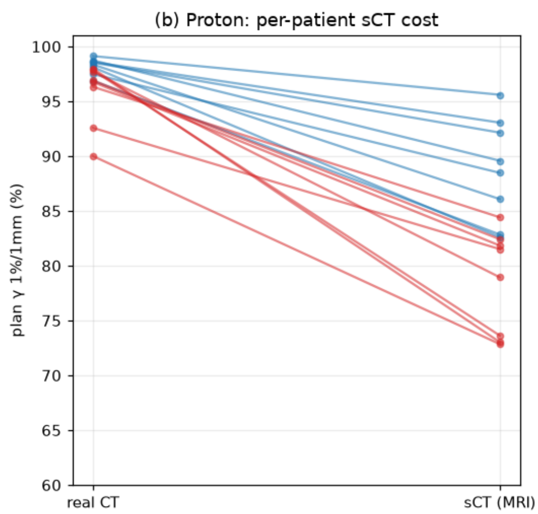

Fig. 4. Per-patient synthesis cost on this task (real CT vs. MR-derived density input, development fold; blue abdomen, red lung): *every* patient loses accuracy, lung most; proton range integrates density. The corresponding photon cost is $\sim 2\times$ smaller (see text).

abdomen-driven trends, exactly where MR-recoverable density lives; but individually within noise at n=16, and both are null on the photon-MRI task by design. We therefore rest the interpretive weight not on these deltas but on the highly significant particle contrast (every patient loses under proton synthesis, $p \ll 0.001$) and on the cross-validated accuracy. Bone/air segmentation fixes and loss re-weighting toward bone likewise did not move gamma (p=0.57), consistent with our diagnosis that the residual synthesis error is spatially unstructured soft-tissue error plus lung, not a fixable segmentation artifact.

Table 3. Proton-MRI method ablations, per-patient paired tests (development fold, $n = 16$, $\gamma 1\%/1\,\text{mm}$).

Comparison	$\Delta\gamma$ (mean±SD)	W/T/L	Wilcoxon p
+WEPL-consistency vs dose-aware baseline	$+0.37 \pm 1.22$	10/0/6	0.35
+density-direct vs +WEPL	$+0.36 \pm 0.84$	8/1/7	0.25
(ref) proton synthesis cost	$\mathbf{-13.3 \pm 6.1}$	16/0/0	$\ll \mathbf{0.001}$

3.4 Decomposing the synthesis error, and what did not fix it

Where does the 8.7-point cost live? Direct decomposition of the deployed synthesis against the paired CT gives three components. *Range*: the WEPL-relevant integrated error corresponds to ~3 mm of range uncertainty where ~1 mm would be needed for CT-grade gamma; the dominant term, matching the R_{80} analysis. *Soft tissue*: ~41% of in-body density error sits in soft tissue and is spatially unstructured (near-random residual after the refiner), i.e. not attributable to any segmentable class. *Interfaces*: bone/air boundary errors exist but are minor contributors. This decomposition predicted, correctly, that several plausible fixes would fail, and we report them as tested negatives: bone-weighted anchor losses ($+0.05\,\gamma$, p=0.57); better lung segmentation (TotalSegmentator-MRI mask, no gamma gain); retraining the synthesis at the native 1×1×3 mm grid (failed our no-loss gate; the apparent "2 mm bottleneck" was illusory); and whole-image synthesis variants (on par, not better). What *did* move the board was none of the above but the intensity-shift-augmented classifier (+3.1; Sec. 2.3): generalization, not in-domain synthesis fidelity, was the binding constraint. A methodological lesson underlying all of this: in-sample validation misled us early (γ 96-level in-sample vs. 76 held-out on our first board submission), and every conclusion above is from held-out or leaderboard measurement.

3.5 Runtime engineering

Proton-MRI shares the Proton-CT deployment stack and adds the synthesis stage (classifier sliding-window + refiner, ~2 s per source image). The first submitted engine reduced runtime from 414 s to 188.8 s per job chiefly through streamed zlib-compressed output with a writer pool (proton jobs write tens of GB on the native grid and are write-bound without overlap). The batched engine then collapsed per-beam WEPL to one orthographic beam's-eye-view cumulative sum (beamlet rays within a beam are exactly parallel; 388× on the WEPL stage), bucket-batched the network forward (exact by same-shape bucketing), and streamed per-beam chunks: measured 110 s per job on the platform at unchanged accuracy. A platform finding worth reporting: the evaluation service starts a fresh process per job, so `torch.compile` pays a ~50 s tax per job and recompiles on each new patient grid (one measured multi-grid job: 238 s compiled vs. 110 s eager); the proton engines therefore ship without compile.

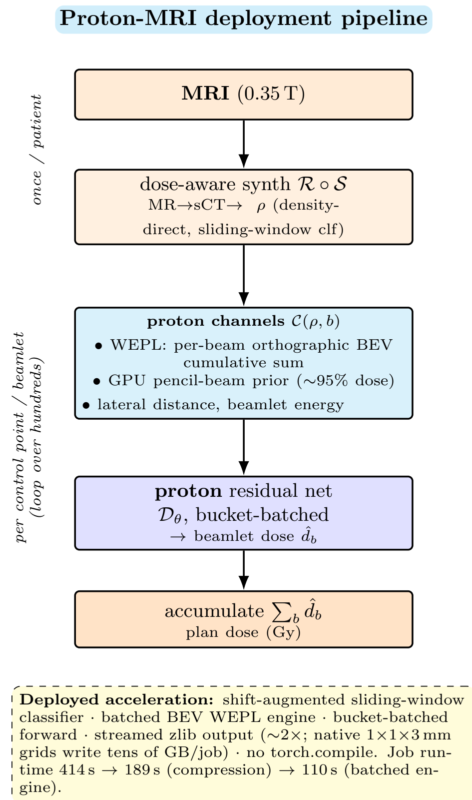

Fig. 5. The deployed Proton-MRI container: per patient, the density stage runs once; the differentiable-channels → residual-net loop runs per control point/beamlet. The callout lists the deployed acceleration measures and their measured effects.

4 Discussion

The central quantitative result of this task is an accuracy cost: MR-only proton dose prediction at 0.35 T costs 8.7 ± 1.9 gamma points against the same pipeline on real CT, it is a *range* effect (6× larger distal-R_{80} shift, concentrated in lung), and it is roughly twice the corresponding photon cost. Dose-aware synthesis, density-direct output and WEPL supervision each push in the right direction but do not close it; nor did better segmentation or bone-weighted losses. We conclude the residual gap is limited primarily by the density information available in 0.35 T bSSFP MR (and by cohort size), not by network capacity, which locates the promising levers at higher field strength, larger MR-only cohorts, and possibly sCT-free direct MR-to-dose learning, rather than at architecture. **Limitations and generalizability.** Single benchmark, single scanner; the hidden-test intensity shift is a preview of cross-device shift, and our shift-augmented

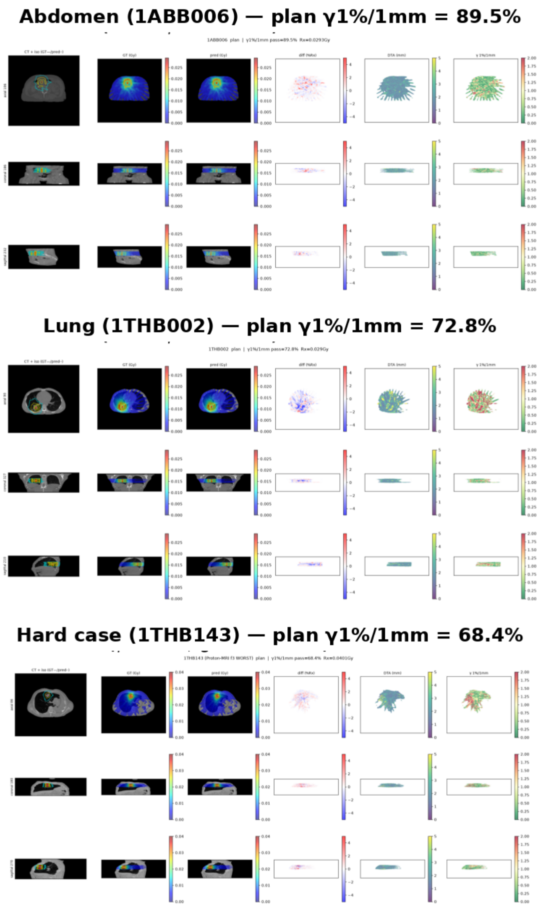

Fig. 6. Qualitative Proton-MRI results on three cases spanning the difficulty range: abdomen (1ABB006, plan $\gamma 1/1 = 89.5$), lung (1THB002, 72.8), and the geometrically hard case 1THB143 (68.4; the worst patient in all four challenge tasks simultaneously, yet largely passing at 3%/3 mm: small-magnitude errors at steep lung–tissue interfaces). Columns: image with GT (solid) and predicted (dashed) isodose, MC dose, predicted dose, difference, distance-to-agreement, local γ1%/1 mm.

classifier recovers only part of it. Ground truth is CT-computed dose mapped to the MR, so registration residuals are label noise. Method-level ablations are single-fold (n=16) and individually non-significant; the particle contrast and the CV accuracy carry the conclusions. For clinical MR-only proton planning, the ceiling analysis offers a usable decision tool: it measures, per-site, how much accuracy the MR-to-density path costs against a CT-based plan.

Why an intermediate density image? The density map is the interface to the physics: WEPL and the pencil-beam prior consume it, and they supply ∼95% of the dose signal that a direct MR-to-dose network would need to relearn from 75 patients. The synthesizer is partially transferred (166 field-matched SynthRAD2025 pairs pretrain the refiner). Note also that the dose-aware density genuinely diverges from an imaging-optimized sCT: optimized through the dose loss and the WEPL-consistency term, it trades image fidelity for range fidelity, and should not be reused as a diagnostic image. Whether an sCT-free direct MR-to-dose model can match this pipeline is untested here and is our first future-work item.

Future work. Our error decomposition dictates the roadmap. (i) *sCT-free direct MR-to-dose*: since the soft-tissue residual is unstructured and every synthesis-side fix we tested was flat, the most promising escape is to stop forcing the information through an explicit density image: learn dose (or a dose correction to a provisional-density prior) directly from the MR, keeping the differentiable physics as scaffolding. (ii) *Range-first data*: higher field strength and larger MR-only cohorts attack the $\sim$3 mm range error at its information source; UTE/ZTE or multi-contrast acquisitions that see bone and lung structure are the acquisition-side lever. (iii) *Uncertainty-aware planning integration*: the per-voxel synthesis uncertainty our anchor framework already estimates could feed robust-plan optimization [5] rather than being discarded, turning a known 2–3 mm range uncertainty into margins instead of errors. (iv) *Generalization as a first-class objective*: the intensity-shift result (the single largest board improvement of our campaign) suggests routine adversarial/physics-based MR appearance augmentation and held-out-center validation should precede any further in-domain accuracy work.

5 Author Contributions (CRediT)

Kai Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Meixu Chen:** Software (deployment and runtime optimization), Investigation (acceleration study), Validation, Writing – review & editing. **Rui Yang:** Conceptualization, Methodology, Writing – review & editing.

6 Other Information

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